Double Descent in Linear Models

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Abstract

This project empirically reproduces double descent, a phenomenon popularized in machine learning. By varying the model dimensionality up to over-parametrized setting, it compares performance of Ordinary least squares and Ridge regression on both a synthetic Gaussian dataset and a real dataset (transformed using Random Fourier features). The findings are consistent with recent literature challenging the conventional view that over-parameterization harms prediction. The results reveal a pronounced increase in test error around the interpolation threshold, followed by a second descent as the model becomes over-parameterized. An expected decrease in test risk for over-parametrized models has been observed with significantly smoother curves for regularized Ridge regression, demonstrating reduced predictive instability. Additional extensions investigate the effect of different noise levels and examine the contributions of bias and variance to prediction error. Overall, the findings illustrate that increasing model dimension does not always result in poorer predictive performance.

Plagiarism Declaration

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1 Introduction

The double descent is a fairly new phenomenon which has been popularized in machine learning by Belkin et al. [1] in 2019. Belkin et al. [1] used Figure 1 to explain this phenomenon in simple terms. The first plot is widely known, showcasing the bias-variance trade-off. It shows that less complex models tend to underfit as the variance is too low and so the bias is high. Increasing the complexity of the model lowers both the train and the test risk, but at some point the test error starts to increase again as the variance increases and bias decreases, meaning the model is overfitting. Reaching that point, we would normally stop and stay at the sweet spot where we minimise the test risk to avoid overfitting. However, modern machine learning has violated this repeatedly using over-parametrised neural networks, which surprisingly do not only have zero training error but also perform greatly, achieving minimal test error. [2]

This is explained in the second part of the Figure 1. Increasing the model complexity over the interpolation threshold (where the number of features is equal to the number of observations and where the test risk is at its maximum) does not only leave training error at zero but leads to a second descent in the test risk which can eventually decrease deeper than in the under-parametrized model.

This report empirically reproduces double descent for linear models with both synthetic and real data. It compares Ordinary least squares (OLS) and Ridge regression with several regularization parameters. The extension compares Ordinary least squares with different noise levels and shows the bias variance decomposition as the model complexity increases.

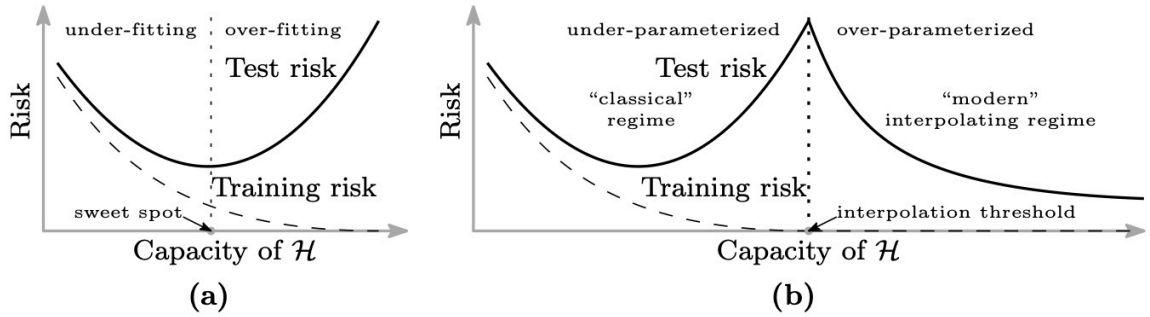


Figure 1: Curves of training and test risk. [Belkin et al.]
a) The classical U-shaped risk curve. b) The double descent risk curve.

2 Theory and Methodology

2.1 Problem Setup

Considering linear regression with n training samples and a model of dimension d. Given a design matrix $X \in \mathbb{R}^{n \times d}$ and a label vector $y \in \mathbb{R}^n$, seeking a weight vector $\hat{\beta} \in \mathbb{R}^d$ that minimizes the mean squared error (MSE):

$$\hat{\beta} = \arg\min_{w \in \mathbb{R}^d} \frac{1}{n} \|y - Xw\|_2^2.$$

2.2 Ordinary Least Squares

Under-parametrized case. When $d \leq n$ the unique OLS solution is

$$\hat{\beta}_{under} = (X^\top X)^{-1} X^\top y.$$

This minimizes the training MSE, for $d < n$, the training error is positive, decreasing and reaching zero as d converges to n.

Over-parametrised case ($d > n$). When $d > n$, the system is not determined properly (infinitely many β satisfy β X = y) the smallest of them is desired. The solution to this optimization problem uses the so-called Gram matrix. [3]

$$\hat{\beta}_{over} = X^\top (X X^\top)^{-1} y.$$

The training error is exactly zero for all $d > n$.

2.3 Ridge Regression

Ridge regression adds an ℓ_2 regularization term to the OLS model providing stabler predictor with lower estimation error (variance):

$$\hat{w}_{\text{ridge}}(\alpha) = \arg\min_w \Big\{ \frac{1}{n} \|y - Xw\|_2^2 + \alpha \|w\|_2^2 \Big\}.$$

The closed-form solution is

$$\hat{w}_{\text{ridge}}(\alpha) = (\alpha I + X^\top X)^{-1} X^\top y,$$

where $\alpha > 0$ ensures that $\alpha I + X^\top X$ is not singular (and hence invertible) for every value of d, including $d > n$. This removes the singularity that occurs at $d = n$ for OLS.

The ℓ_2 penalty shrinks the coefficients towards zero, introducing bias but reducing variance. In the over-parametrized regime, where OLS variance can be very large, even a small α can highly reduce test error.

2.4 The Double Descent Curve

Key intuitions:

- For $d < n$: The model is under-parametrized, there are more data points than features. Standard Ordinary least squares cannot pass through every data point, so training error is positive. The test error follows the classic U-shaped curve.
- For $d = n$: This is the interpolation threshold, around this point $X^\top X$ is nearly singular. The number of features equals the number of training samples. The model fits every single training point perfectly, achieving zero training error. However, fitting noisy data points leads to peak variance and massive test error spike.
- For $d > n$: There are far more features than training data points. Infinitely many solutions can achieve zero training error. Spreading the fitting task across 1,000 small weights instead of 200 larger ones smooths out the model, causing the test error to decrease a second time.

2.5 Bias–Variance Decomposition

The standard decomposition of the expected test MSE is

$$\mathbb{E}\big[(y - \hat{y})^2\big] = \underbrace{\sigma^2}_{\text{noise}} + \underbrace{\big(\mathbb{E}[\hat{y}] - f(x)\big)^2}_{\text{bias}^2} + \underbrace{\mathbb{E}\big[\big(\hat{y} - \mathbb{E}[\hat{y}]\big)^2\big]}_{\text{variance}},$$

3 Experimental Setup

3.1 Synthetic Dataset

The synthetic data is generated according to the Gaussian distribution:

$$x_i \overset{\text{iid}}{\sim} \mathcal{N}(0,1), \quad y_i = x_i\beta + \varepsilon_i, \quad \varepsilon_i \sim \mathcal{N}(0,\sigma^2),$$

with $\beta \in \mathbb{R}^d$ drawn from $\mathcal{N}(0,1)$ and then normalized to unit norm ($\|\beta^*\|_2 = 1$). This keeps the signal strength constant across different values of d, isolating the effect of dimensionality. Using $n_{\text{train}} = 200$, $n_{\text{test}} = 2000$, and default noise level $\sigma = 0.5$, so the noise floor is $\sigma^2 = 0.25$.

The synthetic setting is ideal for studying double descent because the true model is perfectly linear, removing any approximation error from misspecification.

3.2 Real-World Dataset: California Housing

The California Housing dataset pulled with sklearn contains 20,640 observations of Californian census tracts, with 8 features per California districts (e.g. median income, median house age, average rooms) and the median house value as the target. Again $n_{\text{train}} = 200$ and $n_{\text{test}} = 2000$ samples are used.

Random Fourier Features. The original dataset has only 8 features, far fewer than the dimensions d wanted to explore. Random Fourier features [4] helps to lift the input into a d-dimensional feature space that approximates the RBF (Gaussian) kernel $k(x,z) = \exp(-\gamma\|x - z\|^2)$.

Specifically, we sampling $W \in \mathbb{R}^{8\times d}$ with columns drawn i.i.d. from $\mathcal{N}(0, 2\gamma I_8)$ and $b \in \mathbb{R}^d$ uniformly from $[0, 2\pi]$, then define

$$\phi(x) = \sqrt{\frac{2}{d}}\cos(W^\top x + b) \in \mathbb{R}^d.$$

Rahimi and Recht [4] proved that $\mathbb{E}_W[\phi(x)^\top\phi(z)] = k(x,z)$, so the inner product in the feature space approximates the RBF kernel. For this experiment $\gamma = 0.5$ is used.

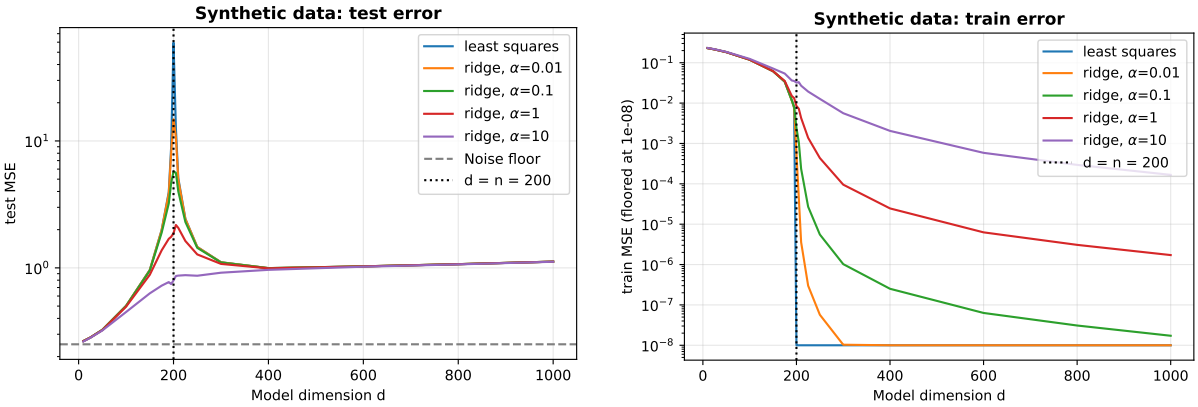
3.3 Model Specifications and Hyperparameters

Parameter	Value
n_{train}	200
n_{test}	2000
Number of trials T	30
Noise std σ	0.5 (synthetic)
RFF bandwidth γ	0.5 (real data)
Dimensions d	10, 25, 50, 100, 150, 175, 190, 195, 200, 205, 210, 225, 250, 300, 400, 600, 800, 1000
Ridge α	0.01, 0.1, 1, 10

Each (d,t) combination uses an independent random seed to generate both the data and the RFF mapping, ensuring trial independence. Results are reported as the *median* over 30 trials to reduce the influence of extreme variance spikes near the interpolation threshold. Train errors are floored at 10^{-8} for visualisation on a log scale.

4 Results

4.1 Synthetic Data



(a) Test MSE vs. model dimension d.

(b) Train MSE vs. model dimension d.

Figure 2: Synthetic dataset: least squares and ridge regression. The vertical dotted line marks the interpolation threshold $d = n = 200$.

Figure 2a shows double descent in the test error for both OLS and Ridge. However, in the under-parametrized regime ($d < 200$), the test error seem to only increase with the model dimension. The initial decrease in test error typically associated with the classical double descent curve is not observed but the sharp interpolation peak at $d = n = 200$ followed by a second descent is present. The spike is smoother for Ridge especially as the regularization parameter increases. In the over-parametrized regime ($d > 200$), the test error decreases monotonically and stabilizes as all models approach each other perfectly.

The training error (Figure 2b) follows the expected decreasing pattern as model complexity increases. For OLS, it approaches zero once the model has enough parameters to interpolate the training observations. Ridge regression shows a similar pattern, although the training error remains higher for larger regularization parameters (α), since the penalty term discourages large coefficients and prevents the model from fitting the training data as closely.

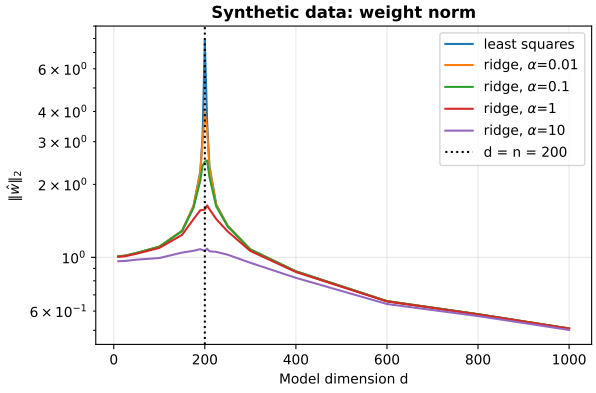
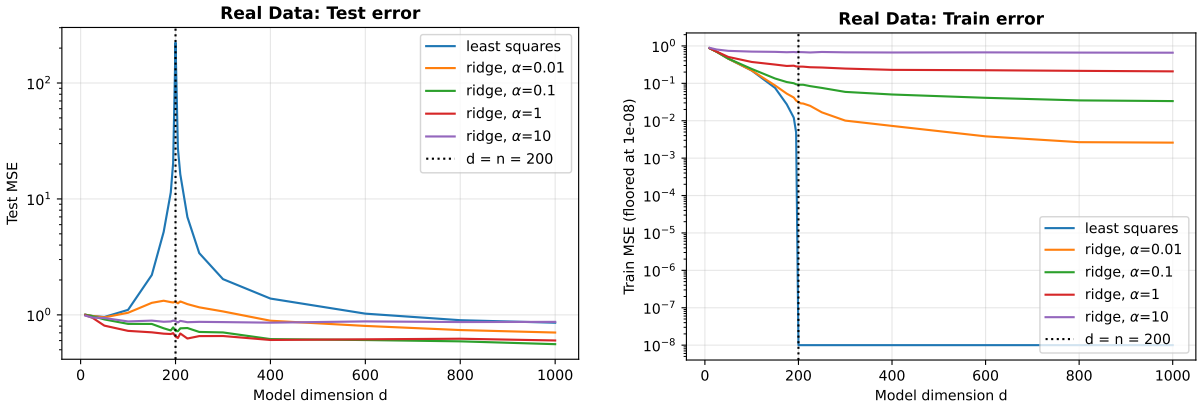


Figure 3: Synthetic dataset: ℓ_2 norm of the fitted weight vector $\|\hat{w}\|_2$.

The coefficient norm displayed in Figure 3 provides further insight into this behavior. For least squares, the norm increases sharply around the interpolation threshold, reflecting the instability of coefficient estimation when the number of parameters approaches the number of observations.

4.2 Real-World Data: California Housing



(a) Test MSE vs. model dimension d.

(b) Train MSE vs. model dimension d.

Figure 4: Real dataset (California Housing with random Fourier features). The vertical dotted line marks interpolation threshold $d = n = 200$.

Figure 4a shows that double descent is also visible on the real dataset, confirming that the phenomenon is not limited to synthetic settings. The spike at $d = n$ is less sharp than in the synthetic case due to the additional noise sources inherent in real data (label noise, model misspecification), but it is clearly present for the OLS model.

In comparison to the results of the synthetic dataset the classical double descent is observed. Initial decline, sudden spike and second decrease are mostly visible for the OLS and Ridge with $\alpha = 0.01$. In this case it is seen clearly that the Ridge does smooth out the curve even with a low hyperparameter. Three of the Ridge models have no visible spike and seem to have almost monotonically decreasing test errors. For all of the models, it can be observed that the second descent eventually does go deeper than the initial one, and so the over-parametrized models are performing better than the under-parametrized ones.

The training error, decreases with model dimension and reaches approximately zero for OLS at the interpolation threshold. As expected, stronger regularization also results in higher training error for Ridge, since the penalty restricts the model's ability to fit the training data.

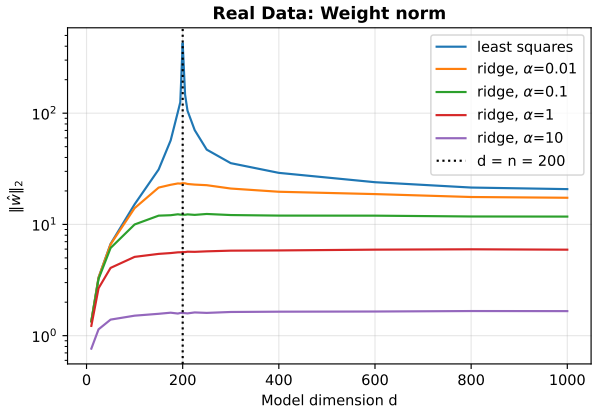


Figure 5: Real dataset: ℓ_2 norm of the fitted weight vector.

Figure 5 shows similar but smoother (especially for Ridge) results compared to the synthetic dataset, OLS exhibits a sharp spike in $\|\hat{w}\|_2$ at $d = n$, while ridge regression keeps it bounded.

5 Extensions

5.1 Effect of Noise

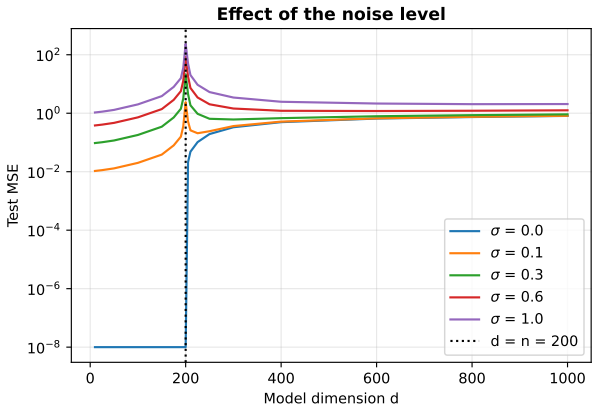


Figure 6: Effect of the noise level σ on the test MSE of OLS.

Figure 6 shows the OLS test error for five noise levels $\sigma \in \{0.0, 0.1, 0.3, 0.6, 1.0\}$.

To investigate the sensitivity of the double-descent phenomenon to noise, the OLS experiment was repeated using different levels of noise. The results show that the shape of the test error curves remains similar across noise levels, with a pronounced increase around the interpolation threshold ($d = n = 200$), followed by a second descent in the over-parametrized regime.

Increasing the noise level leads to higher test MSE across all model dimensions, reflecting the additional uncertainty. In contrast, when no noise is added, test error is approximately zero in the under-parametrized regime, since the underlying data-generating relationship is linear and can be recovered by least squares. Nevertheless, interpolation peak remains even in this noiseless setting.

Overall, these results suggest that the double-descent pattern is not solely a consequence of noisy observations. Rather, it is also related to the relationship between the number of model parameters and the number of training observations. Noise amplifies the prediction error, particularly around the interpolation threshold, but does not fundamentally change the overall pattern observed in this experiment.

5.2 Bias–Variance Decomposition

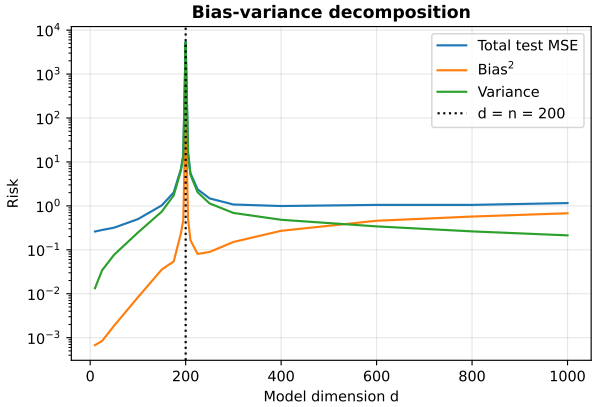


Figure 7: Bias–variance decomposition of the OLS test risk (synthetic data, $\sigma = 0.5$).

Figure 7 plots the bias^2, variance, and total test MSE of OLS as a functions of d.

In the classical bias–variance framework, as the model complexity increases, it is expected to reduce bias while increasing variance (X shaped plot with decreasing bias and increasing variance that cross in the middle as the MSE is at its lowest), resulting in a trade-off between the two components.

The results illustrate that in this case the relationship becomes more complex. As the model dimension approaches the number of training observations, both bias and variance increase, with variance showing a sharp peak at (d=n). Beyond the interpolation threshold, both variance and bias start to decrease as model dimension increases.

After the initial drop bias starts to increase and eventually crosses variance. The underlying reason for the increase in bias in over-parametrized regime is that, the model does not have enough training data points to sample the entire input feature space. Therefore, any predictions made about these unsampled directions represent implicit assumptions of the model.[6] In addition to label noise, nonlinear relationships in the data can introduce an irreducible component of bias, since a linear model cannot fully capture the underlying nonlinear relationship.[7]

This demonstrates that the classical assumption of decreasing bias and increasing variance does not necessarily hold in over-parameterized settings. Nevertheless, the bias–variance decomposition itself remains valid, as both components continue to contribute to the expected prediction error together with irreducible noise.

6 Conclusion and Discussion

Interpolation threshold represents a region of high estimation instability, rather than a point beyond which model performance will continue to worse.

The experiments using both synthetic and real-world data showed that both cases follow the double descent phenomenon revealing interpolation peak around the point where the number of model features equals the number of training observations, followed by a second descent in test error in the over-parameterized regime. Some of the models followed the double descent as expected others satisfied only some of its aspects.

The comparison between Ordinary least squares and Ridge regression highlighted the importance of regularization in controlling estimation instability, as penalising large coefficients substantially reduced the interpolation peak and resulted in more stable test performance and smoother curves, at the cost of higher training error.

The additional noise experiment showed that the general double descent pattern persisted across different noise levels, increasing noise generally inflated prediction errors. Furthermore, the bias–variance decomposition illustrated that the classical decomposition remains valid in over-parameterized settings, although the behavior of its individual components can differ from the conventional bias–variance trade-off.

Despite these findings, the experiments have some limitations. The synthetic data rely on a simplified linear data-generating process with independent normally distributed features, and the underlying coefficient vector changes across model dimensions, limiting direct comparisons between different levels of complexity. The real-world analysis is also based on a relatively small training sample and a random Fourier feature transformation, meaning that the results may depend on the selected transformation parameters. Additionally, only a limited number of regularization parameters and repetitions were considered. Future research could address these limitations by investigating different sample sizes, more complex data-generating processes, systematic selection of regularization parameters through cross-validation and alternative estimation methods. Adding

gradient descent would allow studying how implicit regularization of gradient descent interacts with the double descent phenomenon (see[5]). Despite any findings it cannot be concluded that larger models will perform better, there are many factors that need to be considered. Larger model does not necessarily mean better test performance.[8]

Overall, the results demonstrate that increasing model complexity does not necessarily lead to poorer generalization and that the relationship between model dimension, training sample size, noise and regularization plays an important role in determining predictive performance.

Code availability.
All code is available in the repository at https://github.com/Jarka8/Double_descent_in_linear_models.

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